

Point Operators

- Gamma correction is a nonlinear operation used to adjust the brightness and contrast of digital images by compensating for the way humans perceive light

Mathematically:

$$g(x, y) = T(f(x, y))$$

- $f(x, y) \rightarrow$ input pixel intensity
- $g(x, y) \rightarrow$ output pixel intensity
- $T(\cdot) \rightarrow$ transformation function

Point operators

- **Image Inversion (Negative)**
- **Gamma Correction**
- **Contrast Stretching**
- **Thresholding**
- **Histogram Equalization**

Point Operator rule (always remember):

- Each pixel is processed independently.
- No neighboring pixels are used.
- So we are only changing **pixel brightness values**, not shapes.

Image Inversion (Negative Image)

- **What does “invert” mean?**

Invert means:

- Bright becomes dark
- Dark becomes bright

Like a **photo negative**.



Think in very simple terms

- Imagine brightness values go from **0 to 255**:

0 → black

255 → white

Inversion flips this scale.

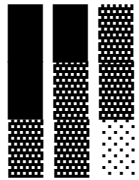
- **Formula:**

$$\text{Output} = 255 - \text{Input}$$

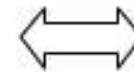
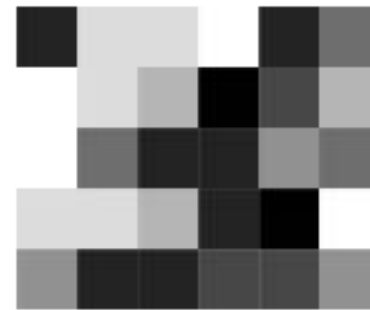
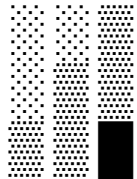
Input Pixel	Output Pixel
0 (black)	255 (white)
50	205
120	135
255 (white)	0 (black)

Visual intuition

- Before



After (inverted):



6	1	1	0	6	4
0	1	2	7	5	2
0	4	6	6	3	4
1	1	2	6	7	0
3	6	6	5	5	3

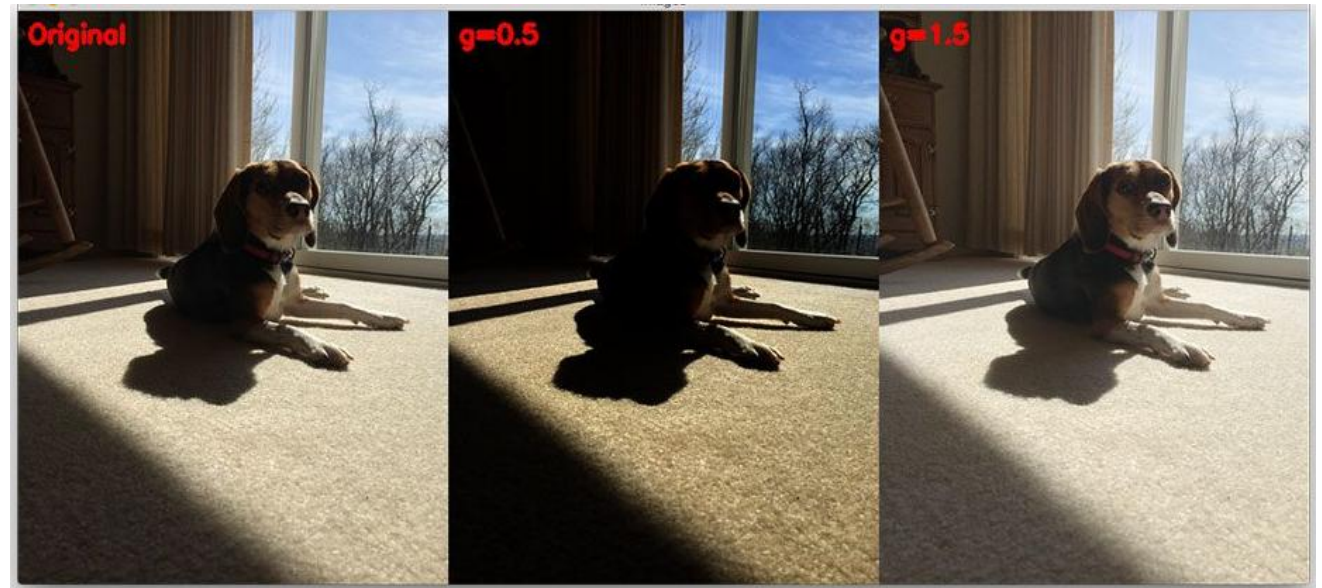
Where is inversion used?

- Medical X-rays (bones become clearer)
- Highlighting hidden details



Gamma Correction

- Gamma correction is a nonlinear operation used to adjust the brightness and contrast of digital images by compensating for the way humans perceive light
- **What problem does gamma solve?**
 - Cameras and screens **do not show brightness linearly**.
 - Gamma correction adjusts brightness **non-linearly** to:
 - brighten dark images
 - darken overly bright images
- **Simple idea**
 - Some images look **too dark**
 - Some look **washed out**
 - Gamma correction fixes this



Gamma Correction

- Formula (conceptual)

$$\text{Output} = (\text{Input})^\gamma$$

- You don't calculate this by hand — computers do it.

➤ Important cases

Case 1: $\gamma < 1$

➡ Dark areas become brighter

Case 2: $\gamma > 1$

➡ Bright areas become darker

Example (conceptual)

Input pixel = 50

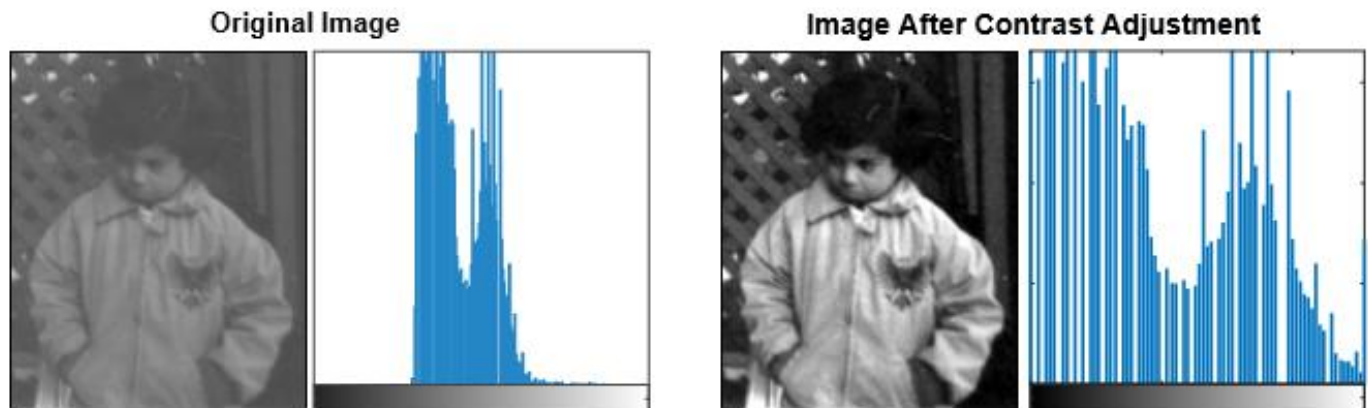
- $\gamma = 0.5 \rightarrow \text{output} \approx \text{brighter}$
- $\gamma = 2 \rightarrow \text{output} \approx \text{darker}$

- **Where is gamma correction used?**

- Cameras
- Mobile phones
- TVs and monitors
- Low-light image enhancement

Contrast Stretching

- Contrast stretching is an image enhancement technique used in digital image processing to **improve the visibility of features by expanding the range of intensity values** in a low-contrast image
- **What is contrast?**
 - Contrast = **difference between dark and bright areas**
- Low contrast image:
 - Everything looks gray
 - No clear details
- **What does contrast stretching do?**
 - It **spreads** pixel values over the full range (0–255).
 - Dark → darker
 - Bright → brighter



- **What problem does it solve?**

- An image may use only a **small range of intensities**, e.g. 50–150 instead of 0–255. Contrast stretching **expands** this range.

- **Formula (Linear Contrast Stretching)**

$$g(x, y) = \frac{f(x, y) - f_{\min}}{f_{\max} - f_{\min}} \times (L - 1)$$

Where:

- $f(x, y)$ = input pixel
- $f_{\min}$ = minimum intensity in image
- $f_{\max}$ = maximum intensity in image
- L = number of gray levels (256 for 8-bit)

- Very Simple Example

$$f_{\min} = 50$$

$$f_{\max} = 150$$

Pixel value $f = 100$

$$g = \frac{100 - 50}{150 - 50} \times 255 = \frac{50}{100} \times 255 = 127$$

Mapping:

Input	Output
50	0
100	127
150	255

Contrast is increased.

Intuition

- Dark values pushed darker
- Bright values pushed brighter
- Middle values spread evenly

Thresholding

- **What problem does it solve?**
 - Separates **object from background** by converting image into **binary**.

Formula (Basic Thresholding)

$$g(x, y) = \begin{cases} 1 & \text{if } f(x, y) \geq T \\ 0 & \text{if } f(x, y) < T \end{cases}$$

- (Or sometimes 255 and 0 instead of 1 and 0)

Where:

T = threshold value



Original Image



Thresholded and segmented Image

Thresholding

Let threshold $T=100$

Input Pixel	Output
40	0 (black)
90	0 (black)
100	1 (white)
160	1 (white)

Visual Meaning

- Pixels brighter than threshold $\rightarrow$ object
- Pixels darker $\rightarrow$ background

Important Note

- Thresholding is very sensitive to lighting.
- That's why advanced methods (Otsu, adaptive thresholding) exist.